Javad Komijani

Curriculum Vitae

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Professional Summary

Computational physicist and scientific machine-learning researcher with 10+ years of experience in lattice gauge theory, Bayesian inference, differentiable computing, and generative modeling. Developer of GPU-accelerated PyTorch frameworks for diffusion models, normalizing flows, and Lie-group-valued systems. Research interests include symmetry-preserving generative models, scientific machine learning, and scalable algorithms for physical simulation.

Education

2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*
- Studied effective field theories, built Bayesian inference pipelines for extracting physical parameters from lattice-QCD simulations, and determined decay constants of D mesons.
- Developed a novel definition for eigenvalues in the context of nonlinear problems.

2009 **MSc in Electrical Engineering**, *University of Tehran*
- Focus: electromagnetic waves and fields, signal processing, statistical analysis.

2006 **BSc in Electrical Engineering**, *University of Tehran*

Professional Experience

2021–2026 **Postdoctoral Researcher**, *ETH Zurich*
- Developed a novel score-matching framework for diffusion models on non-Abelian Lie groups for lattice gauge theory, enabling generative modeling of SU(N) gauge configurations.
- Developed open-source PyTorch extensions for differentiable linear algebra and adjoint-based ODE solvers for Lie-group-valued systems.
- Developed and integrated normalizing-flow modules into the EU interTwin Digital Twin Engine used across multiple scientific computing domains.
- Implemented distributed training infrastructure for large-scale lattice simulations.
- Co-supervised 2 PhD, 6 MSc, 1 BSc, and 1 internship project in scientific machine learning and computational physics.
- Collaborated on international projects in scientific machine learning and high-energy physics.

2019–2020 **Postdoctoral Researcher**, *University of Tehran*
- Performed precision determinations of Standard Model parameters using lattice-QCD simulations and Bayesian statistical inference.

2017–2018 **Postdoctoral Researcher**, *University of Glasgow*
- Computed hadronic form factors and quark condensates using lattice-QCD simulations and Bayesian inference.

2015–2017 **Postdoctoral Researcher**, *Technical University of Munich*
- Performed precise determinations of quark masses and various decay constants, using lattice simulations and Bayesian analysis.

Open-Source Scientific ML Software

lattice_ml **Installable Python package for scientific machine learning**
- Diffusion-based generative modeling on Lie groups and gauge theories.
- Adjoint-based differentiable ODE solvers for scalar and Lie-group-valued systems.
- Differentiable SVD and eigendecomposition of unitary matrices.

normflow Installable Python package for normalizing flows on Lie groups and lattice gauge theory, including symmetry-preserving transformations and flow-based sampling methods.

interTwin Contributed normalizing-flow-based ML modules to the EU interTwin Digital Twin Engine.

Selected Projects

SU(N) Diffusion Developed one of the first score-based diffusion frameworks on compact Lie groups, enabling generative modeling of lattice gauge configurations with exact symmetry preservation. Applied to generative modeling of SU(3) gauge configurations in two and four dimensions.

Quark Masses Proposed and implemented a method to calculate heavy quark masses; led to precise determination of the up-, down-, strange-, charm-, and bottom-quark masses using the MILC highly improved staggered-quark ensembles.

MRS Mass Introduced the minimal-renormalon-subtracted (MRS) mass scheme for precision determinations of heavy-quark masses and hadronic observables.

Eigenvalues Developed a generalized eigenvalue framework for nonlinear operators using properties of solutions to the Schrödinger equation.

Technical Skills

ML/AI PyTorch, Diffusion Models, Normalizing Flows, Generative Modeling, Bayesian Inference, Score Matching, Distributed Training.

Programming Python, C++, Cython, Bash.

Scientific NumPy, SciPy, Pandas, Monte Carlo Methods, Differentiable ODE Solvers.

Infrastructure Linux, Git, HPC, CUDA, MPI, Supercomputing Environments (CSCS).

Mathematics Differential Equations, Linear Algebra, Optimization, Statistical Modeling.

Co-supervision and Mentoring

PhD
- Octavio Vega (2024–present) — Group-equivariant diffusion models for lattice field theory.
- Ben Izett (2026) — Fourier acceleration for SU(N) lattice gauge theory.

MSc
- Lara Turgut (2025–2026) — Gauge-equivariant diffusion models for lattice field theory.
- Ahmed Chahlaoui (2024–2025) — Autoregressive models for lattice configurations.
- Antoine Misery (2024) — Continuous-time generative models for lattice fields.
- Flavia Giehr (2024) — Continuous normalizing flows with adjoint methods.
- Klemen Kersic (2023) — Rational-quadratic splines for normalizing flows.
- Lea Segner (2023, co-supervised with J. Pinto) — Simulated tempering for topological freezing.

BSc
- Nicolas Müller (2025) — Path integral formulation of quantum mechanics and Monte Carlo Simulations.

Intern
- Elias Nyholm (2023) — CNN architectures in arbitrary dimensions.

Teaching Experience

ETH Zurich
- **Substitute lecturer & head teaching assistant:** Quantum Mechanics II (Spring 2024), Quantum Mechanics I (Fall 2024).
- **Proseminar mentoring and supervision:** General Relativity (Spring 2025), Riemann Surfaces in Mathematical Physics (Spring 2023), Systems Out of Equilibrium (Fall 2022), Quantum Information (Spring 2022), Theoretical Physics (Fall 2021).

UT
- **Organizer and lecturer:** Workshop on lattice field theory for students from several universities in Tehran (Fall 2019).

UG
- **Teaching assistant:** C Programming under Linux (Spring 2018).

TUM
- **Teaching assistant:** Elementary Particle Physics (Spring 2016) and Quantum Field Theory (Fall 2015).

Washington University in St. Louis
- **Teaching assistant:** Statistical Mechanics (Spring 2015), Quantum Mechanics (Fall 2013), Physics I (Fall 2012), Physics II (Spring 2011), Special Relativity (Spring 2011).

Invited Presentations

ECT* 2026 — Bridging machine learning and statistical field theory through generative and diffusion processes, ECT* in Trento, Italy, 7–11 Sep, 2026.

ECT* 2023 — Machine learning for lattice field theory and beyond, ECT* in Trento, Italy, 26–30 Jun, 2023.

ETH 2022 — Efficient simulations on GPU hardware, ETH Zurich, 24–27 Oct, 2022.

$g-2$ 2022 — Fifth Plenary Workshop of the Muon $g-2$ Theory Initiative, University of Edinburgh, 5–9 Sep, 2022.

CKM 2018 — 10th International Workshop on the CKM Unitarity Triangle, Heidelberg, Germany, 17–21 Sep, 2018.

Lattice 2018 — 36th Annual International Symposium on Lattice Field Theory, Michigan State University, 22–28 Jul, 2018.

Quarkonium 2017 — 12th International Workshop on Heavy Quarkonium hosted at Peking University, Beijing, China, 6–10 Nov, 2017.

Research Grants & Projects

CSCS 2022–2025 — Co-Principal Investigator on a three-year CSCS project measuring isospin-breaking effects in hadronic vacuum polarization.

itwinai 2023–2025 — Scientific contributor to the EU Horizon Europe interTwin project, developing machine-learning modules for lattice-QCD simulations.

Professional Service

Refereeing — Referee for Physical Review D and Physical Review Letters (13+ reviews since 2016).

Organization — **Co-organizer of international and local workshops**
- ZPW 2026 — Machine Learning Techniques for Particle Physics.
- QCD 2025 — Symposium on QCD Factorization.
- ML Lattice 2025 — Machine learning approaches in Lattice QCD.
- UT 2019 — Lattice QCD workshop.
- TUM 2017 — EFT and Lattice Field Theory Summer School.

Publications

Research Metrics
- 26 peer-reviewed publications
- h-index: 19
- 3000+ citations (Google Scholar)

Selected Publications:

[1] J. Komijani, M. Marinkovic and L. Turgut, "Diffusion model for SU(N) gauge theories," [arXiv:2605.06134].

[2] O. Vega, J. Komijani, A. El-Khadra and M. Marinkovic, "Group-Equivariant Diffusion Models for Lattice Field Theory," [arXiv:2510.26081].

[3] A. Manzi *et al.*, "interTwin: Advancing Scientific Digital Twins through AI, Federated Computing and Data", *Future Generation Computer Systems, 179, 108312 (2026)*.

[4] R. Aliberti *et al.*, "The anomalous magnetic moment of the muon in the Standard Model: an update," *Phys. Rept. 1143, 1-158 (2025)* [arXiv:2505.21476].

[5] J. Komijani, "First-order nonlinear eigenvalue problems involving functions of a general oscillatory behavior," *J. Phys. A: Math. and Theor. 54, 465202 (2021)* [arXiv:2107.02475].

[6] A. Bazavov *et al.*, "Up-, down-, strange-, charm-, and bottom-quark masses from four-flavor lattice QCD," *Phys. Rev. D 98, 054517 (2018)* [arXiv:1802.04248].

[7] N. Brambilla , J. Komijani, A.S. Kronfeld, A. Vairo, "Relations between Heavy-light Meson and Quark Masses," *Phys. Rev. D 97, 034503 (2018)* [arXiv:1712.04983].

[8] J. Komijani, "A discussion on leading renormalon in the pole mass," *JHEP 1708, 062 (2017)* [arXiv:1701.00347].

[9] C.M. Bender, A. Fring and J. Komijani, "Nonlinear Eigenvalue Problems," *J. Phys. A: Math. and Theor. 47, 235204 (2014)* [arXiv:1401.6161].

[10] C. Bernard and J. Komijani, "Chiral Perturbation Theory for All-Staggered Heavy-Light Mesons," *Phys. Rev. D 88, 094017 (2013)* [arXiv:1309.4533].

Full list of peer-reviewed publications: available here.